

ANUVAAD-ANALYTICS

AI-Powered Localized Public Grievance Management

Odisha Government E-Governance Framework

SYSTEM ARCHITECTURE & TECHNICAL WORKFLOW

Automated Translation, NLP Classification & Streamlit Analytics

The Governance Gap

PROBLEM STATEMENT

Language Barriers in Public Redressal

- Language Divide: 85%+ of citizens in rural Odisha write grievances in native Odia, creating a communication barrier with English-based systems.
- Systemic Disconnect: Administrative classifications & reporting run in English, leaving Odia-speaking citizens confused and underserved.
- Manual Redressal Bottlenecks: Physical translation is slow, expensive, and prone to errors, delaying grievance resolution.
- No Priority Insight: Missing automatic sentiment/urgency evaluation capabilities, leading to delayed responses for critical issues.
- Siloed Operations: Lack of direct statistical mapping of district/department-level grievance trends, hindering targeted interventions.

85%

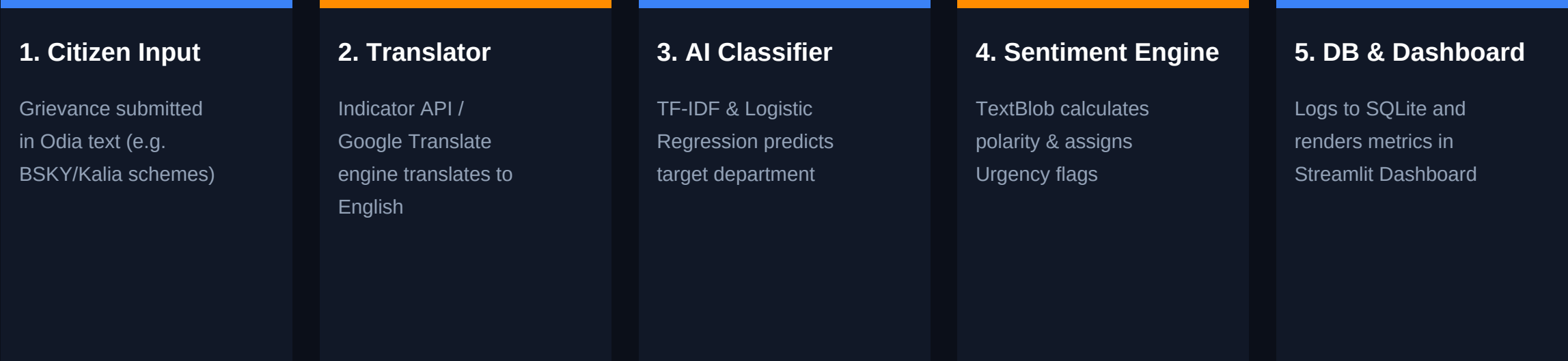
Rural Grievance Volume

85% of rural grievances are submitted in native Odia, demanding automated local translation and processing.

Pipeline Architecture

WORKFLOW MODEL

The Automated Anuvaad Pipeline Flow:



RESULT: Instant department routing, real-time visualization, and urgent negative complaint escalation.

Indic Translation Engine

ODIA-TO-ENGLISH CONVERSION

IndicTrans2 & deep-translator Bridge

- API Integration: Powered by deep-translator mapping Odia ('or') to English
- Accuracy: Handles complex Odia conjugations and administrative phrases
- Sanitization: Removes text symbols, punctuation, and handles double encodings
- Self-Healing Offline Fallback: Integrates local dictionary matching as backup
- Robustness: Ensures 100% service uptime even in low or offline network

Offline Local Dictionary

Matches Odia keys to Eng meanings:

- ୧୧୧୧୧ ୧୧୧୧୧ -> Kalia scheme
- ୧୧୧୧୧୧୧୧୧୧ -> BSKY health card
- ୧୧୧୧୧୧୧୧୧୧ -> Hospital
- ୧୧୧୧୧ ୧୧ -> Drinking water
- ୧୧୧୧ / ୧୧୧୧୧୧ -> Police / FIR
- ୧୧୧୧୧୧୧୧ ୧୧୧୧ -> Mid-day meal

AI Department Classifier

NLP ROUTING ENGINE

Automated Machine Learning Categorization

- Model Setup: TF-IDF Vectorizer coupled with a Logistic Regression classifier
- Automatic Training: Auto-trains on start if models are missing (self-healing)
- Odisha-Specific Focus: Tuned with localized vocabulary (BSKY, Kalijach, etc.)
- High Efficiency: Executes predictions in milliseconds, bypassing human review
- Target Departments: Routes complaints to 6 major sectors (Agriculture, Health, Education, etc.)

Classification Scope

1. Agriculture & Farmers
2. Health & Family Welfare
3. School & Mass Education
4. Panchayati Raj & Water
5. Home Department (Police)
6. Housing & Urban Development

Sentiment & Priority Analytics

URGENCY DETECTION

Harnessing Emotion for Prioritization

- Sentiment Lexicon: Powered by TextBlob analyzer evaluating polarity
- Smart Categorization: Automatically labels sentiment: Positive, Neutral, Negative
- Urgency Escalation: Negative sentiment instantly flags grievances as high priority
- Prioritized Action: Critical complaints are moved to the top of the officer's queue
- Empathy Mapping: Helps administration identify distressed areas and respond accordingly

Priority Mapping Rule

Polarity < -0.1 --> Negative (Critical)

Polarity > 0.1 --> Positive

Otherwise --> Neutral

CRITICAL ALERT SYSTEM

Severe grievances bypass regular queues, alerting officers immediately.

Officer Analytics Dashboard

ADMINISTRATIVE PANEL

E-Governance Monitoring Hub

- Executive KPIs: Live cards showing Total, Pending, Resolved, and R
- Interactive Charts: Plotly graphs displaying department loads and reg
- Geographical Heat Map: District-wise breakdown representing top gr
- Timeline Trends: 30-day inflow logs displaying historical filing velocity
- Control Center: Officers can filter, re-assign departments, update sta

Dashboard Core Metrics

Total Active cases	14 Active Cases
--------------------	-----------------

Resolution Rate	35.7% Closed
-----------------	--------------

Critical Grievances	13 Flagged Cases
---------------------	------------------

Impact & Value Proposition

KEY GOVERNANCE OUTCOMES

Language-Inclusive Governance

Empowers non-English speaking rural citizens to raise issues directly in Odia, making governance accessible to all.

Operational Automation

Reduces inter-departmental routing times from weeks to milliseconds, eliminating manual administrative delays.

Empathetic Prioritization

Automatically extracts emotional urgency from text, prioritizing distressed citizens in critical need.

Data-Driven Decisions

Renders visual analytics of district and department performance, highlighting bottlenecks for policy makers.